Code-Layout, Readability and Reusability

Date	
Team ID	
Project Name	A Comprehensive measure of well-being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	yes	Proper 4-space indentation followed consistently.
2	Proper File Structure	Files and folders are logically organized	yes	Project uses separate folders for model,data,templates and application code.
3	Meaningful Variable Names	Variables reflect their purpose clearly	yes	Variables like MODEL_PATH, FEATURES, COUNTRY_LOOKUP, predicted_score are meaningful.
4	Function / Method Names	Functions are descriptively named	yes	Functions such as validate_inputs(), classify_hdi(), predict clearly describe their purpose.
5	Code Comments	Inline and block comments explain logic	yes	Important sections include comments explaining work flow and functionality
6	Modular Design	Code is split into reusable functions/modules	yes	Validation, classification, routing, and prediction are implemented as separate reusable functions
7	No Redundant Code	Duplicate or unused code is removed	yes	No unnecessary duplicate logic observed.
8	Error Handling	Exceptions and errors are handled gracefully	partial	Input validation is implemented, but model loading (pickle) and file reading(CSV) lack try-except handling.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)

1	Validate _inputs()	Python (flask)	Input validation	High
2	Classify_hdi()	Python	HDI category prediction	High
3	County_data()	Flask API	Auto-fill country values	Medium
4	Predict()	Flask	Prediction processsing	High
5	COUNTRY_LOOKUP()	Pandas dictionary	Country data retrieval	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5/5	Well-organised flask application with clear seprration of responsibilities.
Readability	5/5	Descriptive variable names, functions and comments improve readability.
Reusability	5/5	Functions are modular and reusable across the application.
Documentation / Comments	4/5	Good documentation ;additional comments for complex logic would improve further.
Overall Score	4.8/5	Clean, modular, and maintainable code following good programming practices.